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(54) **INFORMATION PROCESSING SYSTEM**

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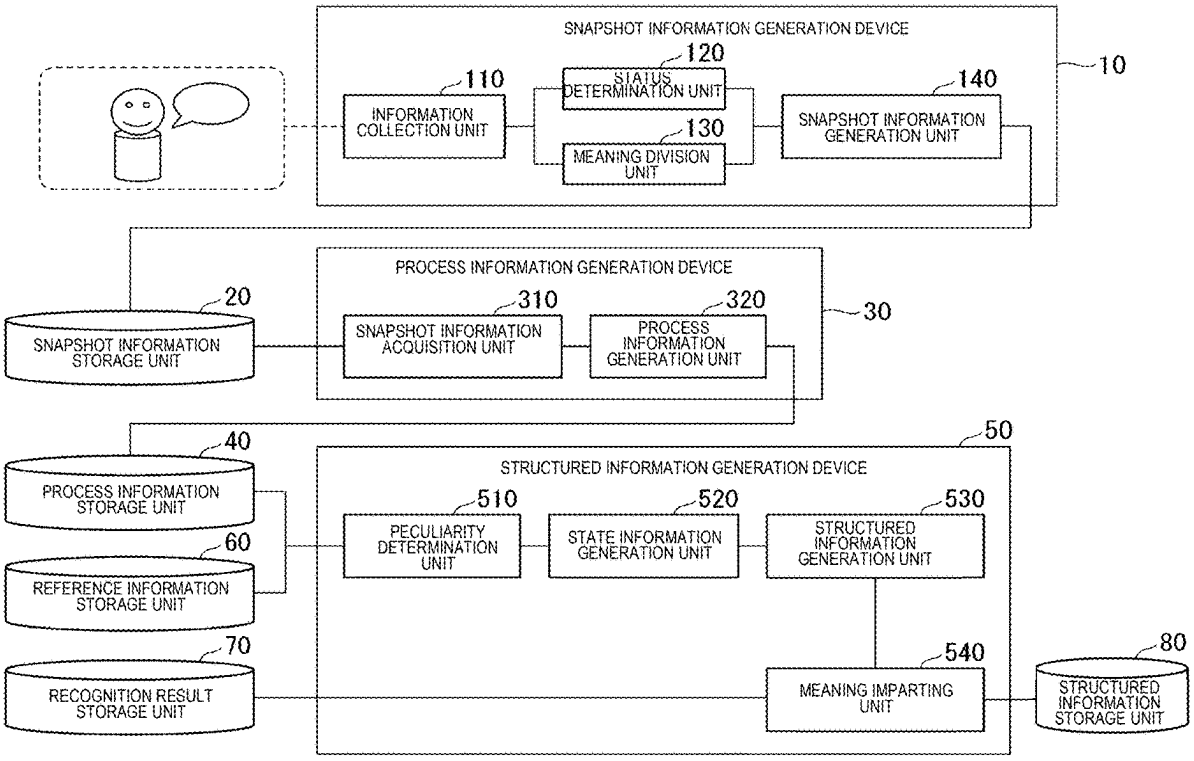
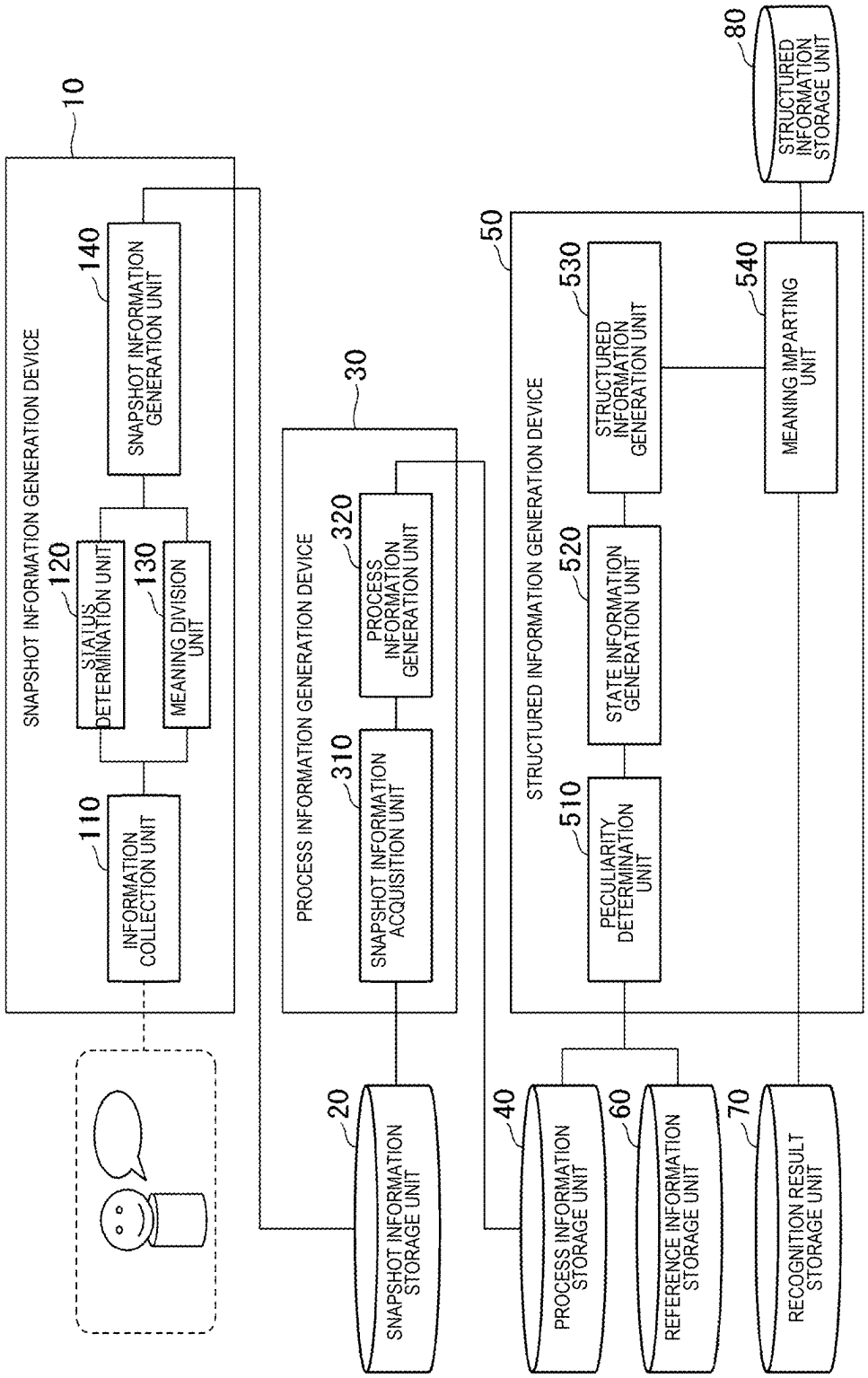
(57) **ABSTRACT**

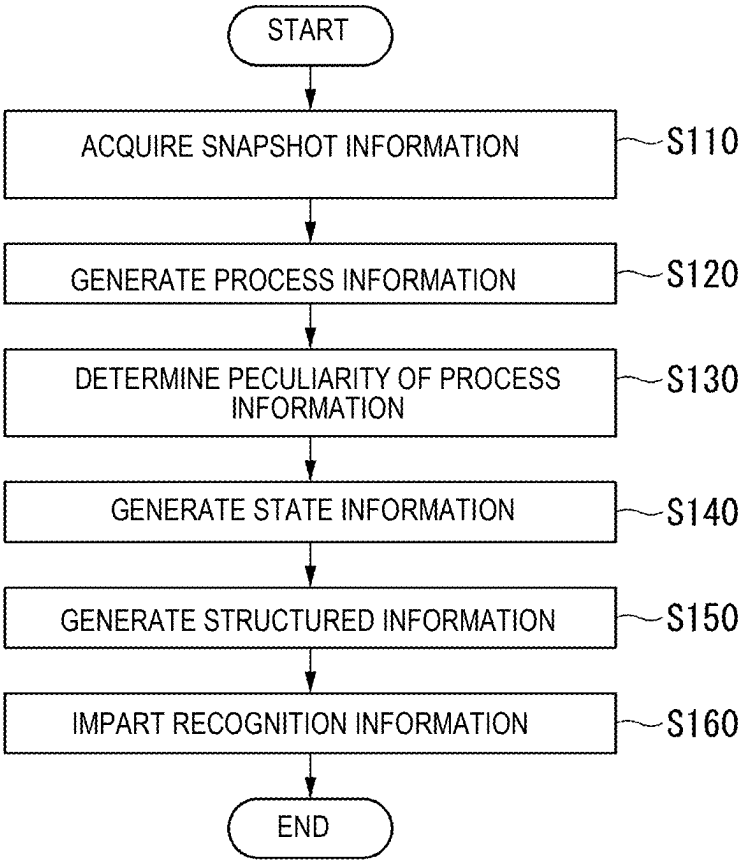
An information processing system includes a snapshot information acquisition unit configured to acquire snapshot information in which unit meaning information and situation information are associated with each other, the unit meaning information being obtained by dividing information indicating a state of a subject into a meaningful unit, the situation information including at least timing information indicating a timing at which the state indicated by the unit meaning information occurs and identification information for identifying the subject; and a process information generation unit configured to generate process information in which the snapshot information is arranged in a time series for each of the subjects based on the timing information.



[FIG. 1]


10
SNAPSHOT INFORMATION GENERATION DEVICE
110
INFORMATION COLLECTION UNIT
120
STATUS DETERMINATION UNIT
130
MEANING DIVISION UNIT
140
SNAPSHOT INFORMATION GENERATION UNIT
30
PROCESS INFORMATION GENERATION DEVICE
310
SNAPSHOT INFORMATION ACQUISITION UNIT
320
PROCESS INFORMATION GENERATION UNIT
50
STRUCTURED INFORMATION GENERATION DEVICE
510
PECULIARITY DETERMINATION UNIT
520
STATE INFORMATION GENERATION UNIT
530
STRUCTURED INFORMATION GENERATION UNIT
540
MEANING IMPARTING UNIT
80
STRUCTURED INFORMATION STORAGE UNIT
20
SNAPSHOT INFORMATION STORAGE UNIT
40
PROCESS INFORMATION STORAGE UNIT
60
REFERENCE INFORMATION STORAGE UNIT
70
RECOGNITION RESULT STORAGE UNIT

[FIG. 2]


START
ACQUIRE SNAPSHOT INFORMATION
S110
GENERATE PROCESS INFORMATION
S120
DETERMINE PECULIARITY OF PROCESS INFORMATION
S130
GENERATE STATE INFORMATION
S140
GENERATE STRUCTURED INFORMATION
S150
IMPART RECOGNITION INFORMATION
S160
END

[FIG. 3]

SSD

SNAPSHOT INFORMATION ID	SITUATION INFORMATION			UNIT MEANING INFORMATION
	SUBJECT IDENTIFICATION INFORMATION	TIMING INFORMATION	···	
SS11	U01	T01	···	M01
SS12	U01	T02	···	M02
SS13	U01	T03	···	M03
···	···	···	···	···
SS21	U02	T04	···	M04
SS22	U02	T05	···	M05
SS23	U02	T06	···	M06
···	···	···	···	···
SS31	U03	T07	···	M07
SS32	U03	T08	···	M08
···	···	···	···	···

[FIG. 4]

(A)

PD1 (PD)

PROCESS INFORMATION ID	SUBJECT IDENTIFICATION INFORMATION	TIMING INFORMATION	UNIT MEANING INFORMATION
PD01	U01	T01	M01
		T02	M02
		T03	M03
		...	...

(B)

PD2 (PD)

PROCESS INFORMATION ID	SUBJECT IDENTIFICATION INFORMATION	TIMING INFORMATION	UNIT MEANING INFORMATION
PD02	U02	T04	M04
		T05	M05
		T06	M06
		...	...

(C)

PD3 (PD)

PROCESS INFORMATION ID	SUBJECT IDENTIFICATION INFORMATION	TIMING INFORMATION	UNIT MEANING INFORMATION
PD03	U03	T07	M07
		T08	M08
		...	...

[FIG. 5]

(A)

CD1 (CD)

STATE INFORMATION ID	TIMING INFORMATION	UNIT MEANING INFORMATION
CD01	T01	M01
	T03	M03

(B)

CD2 (CD)

STATE INFORMATION ID	TIMING INFORMATION	UNIT MEANING INFORMATION
CD02	T04	M04
	T05	M05

(C)

CD3 (CD)

STATE INFORMATION ID	TIMING INFORMATION	UNIT MEANING INFORMATION
CD03	T07	M07
	T08	M08

[FIG. 6]

(A)

MS1 (MS)

STRUCTURED INFORMATION ID	TIMING INFORMATION	UNIT MEANING INFORMATION
MS01	T01	M01
	T03	M03
	T04	M04
	T05	M05
	T07	M07
	T08	M08

(B)

MS2 (MS)

STRUCTURED INFORMATION ID	TIMING INFORMATION	UNIT MEANING INFORMATION
MS02	T01	M01
	T03	M03
	T04	M04
	T05	M05

(C)

MS3 (MS)

STRUCTURED INFORMATION ID	TIMING INFORMATION	UNIT MEANING INFORMATION
MS03	T07	M07
	T08	M08

[FIG. 7]

(A)

IMS1 (IMS)

STRUCTURED INFORMATION ID	TIMING INFORMATION	UNIT MEANING INFORMATION	RECOGNITION INFORMATION
MS01	T01	M01	DIAGNOSIS NAME A
	T03	M03	
	T04	M04	
	T05	M05	DIAGNOSIS NAME B
	T07	M07	
	T08	M08	

(B)

IMS2 (IMS)

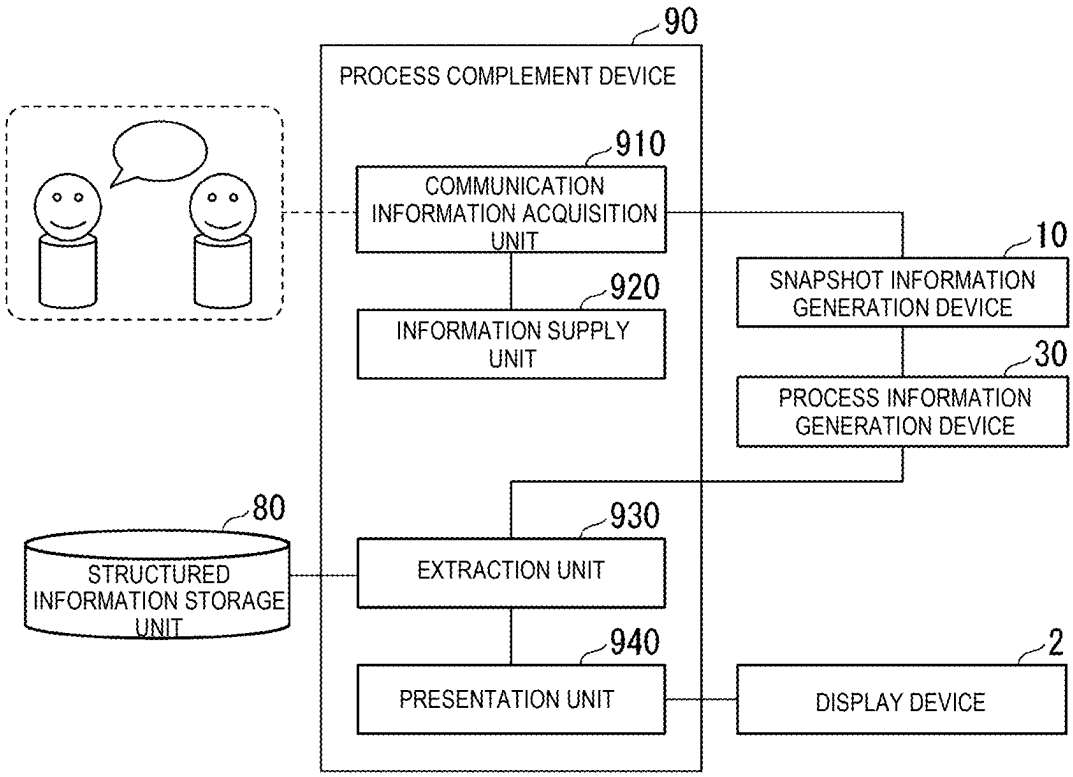
STRUCTURED INFORMATION ID	TIMING INFORMATION	UNIT MEANING INFORMATION	RECOGNITION INFORMATION
MS02	T01	M01	DIAGNOSIS NAME C
	T03	M03	
	T04	M04	
	T05	M05	

(C)

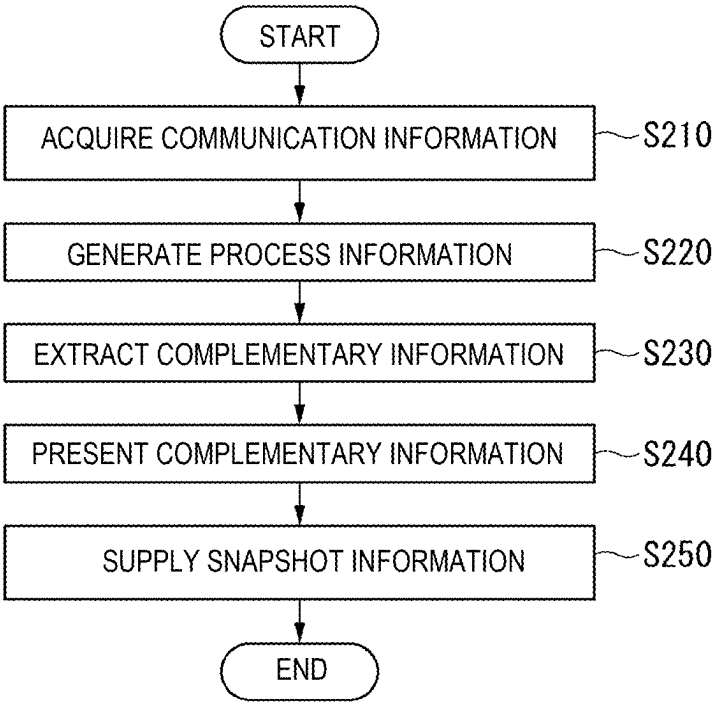
IMS3 (IMS)

STRUCTURED INFORMATION ID	TIMING INFORMATION	UNIT MEANING INFORMATION	RECOGNITION INFORMATION
MS03	T07	M07	DIAGNOSIS NAME D
	T08	M08	

[FIG. 8]


90
PROCESS COMPLEMENT DEVICE
910
COMMUNICATION INFORMATION ACQUISITION UNIT
920
INFORMATION SUPPLY UNIT
10
SNAPSHOT INFORMATION GENERATION DEVICE
30
PROCESS INFORMATION GENERATION DEVICE
80
STRUCTURED INFORMATION STORAGE UNIT
930
EXTRACTION UNIT
940
PRESENTATION UNIT
2
DISPLAY DEVICE

[FIG. 9]


START
ACQUIRE COMMUNICATION INFORMATION
S210
GENERATE PROCESS INFORMATION
S220
EXTRACT COMPLEMENTARY INFORMATION
S230
PRESENT COMPLEMENTARY INFORMATION
S240
SUPPLY SNAPSHOT INFORMATION
S250
END

INFORMATION PROCESSING SYSTEM

TECHNICAL FIELD

[0001] The present invention information processing system.

BACKGROUND ART

[0002] In the related art, a technique for generating a database for work support based on a definition regarding a meaning of information by a human recognition is disclosed (for example, refer to Patent Literature 1).

CITATION LIST

Patent Literature

[0003] Patent Literature 1: JP2020-140604A.

SUMMARY OF INVENTION

Technical Problem

[0004] However, in the technique in the related art as described above, in a case where the database is generated based on the definition by the human recognition, there is a problem in that work of imparting the human recognition to information is necessary, and an enormous number of man-hours are required.

[0005] The present invention has been made to solve the above problem, and an object of the present invention is to provide an information processing system capable of generating structured information based on an enormous amount of snapshot information to improve accuracy of the structured information.

Solution to Problem

[0006] An embodiment of the present invention relates to an information processing system including a snapshot information acquisition unit configured to acquire snapshot information in which unit meaning information and situation information are associated with each other, the unit meaning information being obtained by dividing information indicating a state of a subject into a meaningful unit, the situation information including at least timing information indicating a timing at which the state indicated by the unit meaning information occurs and identification information for identifying the subject; and a process information generation unit configured to generate process information in which the snapshot information is arranged in a time series for each of the subjects based on the timing information.

[0007] The above information processing system according to the embodiment of the present invention further includes a peculiarity determination unit configured to determine a peculiarity [of a part of a time series] of the process information generated by the process information generation unit based on the process information and reference information indicating a predetermined reference; a state information generation unit configured to classify the process information for groups of the peculiarity determined by the peculiarity determination unit to generate state information for each of the groups; and a structured information generation unit configured to generate structured information by combining the state information in different groups in the generated state information.

[0008] The above information processing system according to the embodiment of the present invention further includes a meaning imparting unit configured to impart a meaning to the state indicated by the structured information.

[0009] The above information processing system according to the embodiment of the present invention further includes a communication information acquisition unit configured to acquire communication information indicating a content of communication between users; an extraction unit configured to extract, based on the acquired communication information and the structured information, information, which is required to be complemented based on the structured information, in a time series of the communication as complementary information; and a presentation unit configured to present the extracted complementary information to the users.

[0010] The above information processing system according to the embodiment of the present invention further includes an information supply t configured to supply, to the snapshot information acquisition unit, information indicating a state of a subject obtained as a result of the presentation of the complementary information presented by the presentation unit.

[0011] An embodiment of the present invention relates to an information processing system including: a process information acquisition unit configured to acquire process information in which snapshot information, in which unit meaning information obtained by dividing information indicating a state of a subject into a meaningful unit and situation information are associated with each other, is arranged in a time series for each of the subjects based on timing information indicating a timing at which the state indicated by the unit meaning information occurs, the situation information including at least timing information and identification information for identifying the subject; a peculiarity determination unit configured to determine a peculiarity of the acquired process information based on the process information and reference information indicating a predetermined reference; a state information generation unit configured to classify the process information for groups of the peculiarity determined by the peculiarity determination unit to generate state information for each of the groups; and a structured information generation unit configured to generate structured information by combining the state information in different groups in the generated state information.

Advantageous Effects of Invention

[0012] According to the present invention, it is possible to generate structured information based on an enormous amount of snapshot information to improve accuracy of the structured information.

BRIEF DESCRIPTION OF DRAWINGS

[0013] FIG. **1** is a diagram showing an example of a configuration of an information processing system according to an embodiment.

[0014] FIG. **2** is a diagram showing an example of a processing flow in a recognition mode of the information processing system according to the present embodiment.

[0015] FIG. **3** is a diagram showing an example of a structure of snapshot information according to the present embodiment.

[0016] FIG. **4** is a diagram showing an example of a structure of process information according to the present embodiment.

[0017] FIG. **5** is a diagram showing an example of a structure of state information according to the present embodiment.

[0018] FIG. **6** is a diagram showing an example of the structure of structured information according to the present embodiment.

[0019] FIG. **7** is a diagram showing an example of a structure of meaning matching structured information according to the present embodiment.

[0020] FIG. **8** is a diagram showing an example of a configuration of the information processing system in an application mode according to the present embodiment.

[0021] FIG. **9** is a diagram showing an example of a processing flow in the application mode according to the present embodiment.

DESCRIPTION OF EMBODIMENTS

Overview of Information Processing System 1

[0022] An information processing system **1** of the present embodiment converts human thought into data in a computer-recognizable form, classifies a thought process (process), and accumulates the classified thought process in a database. The information processing system **1** supports human thought by complementing the human thought process with information accumulated in the database.

[0023] The information processing system **1** can be used in various fields. In the present embodiment, a case in which the information processing is system **1** used as s an examination navigation system at the time of examination of a patient by a doctor is described as an example. In this example, a user of the information processing system **1** includes a doctor and a patient.

[0024] Hereinafter, the following description is divided into a process (recognition mode) in which a computer of the information processing system **1** performs recognition and a process (application mode) in which the computer applies a recognition result. First, a configuration of the information processing system **1** in the recognition mode will be described with reference to FIG. **1**.

[0025] The recognition mode may be referred to as a stage (learning stage) of accumulating information in a database. The application mode may be referred to as a stage (use stage) of using information accumulated in the database.

Configuration of Information Processing System 1 in Recognition Mode

[0026] FIG. **1** is a diagram showing an example of the configuration of the information processing system **1** according to the present embodiment.

[0027] The information processing system **1** includes a snapshot information generation device **10**, a process information generation device **30**, and a structured information generation device **50**.

[0028] Each of the devices includes an arithmetic circuit such as a central processing unit (CPU) and a storage device (both not shown), and operates according to a program built in the storage device (or a program supplied from the outside). Each of the devices provides various functions in cooperation with the arithmetic circuit and the storage device. Among the functions provided by the devices, a functional unit provided according to a software configuration is referred to as a software functional unit, and a functional unit provided according to a hardware configuration is also referred to as a hardware functional unit. In the following description, the software functional unit and the hardware functional unit are simply referred to as a functional unit.

[0029] The snapshot information generation device **10**, the process information generation device **30**, and the structured information generation device **50** may be integrated and implemented in a single computer device or may be divided and implemented in a plurality of computer devices. The snapshot information generation device **10**, the process information generation device **30**, and the structured information generation device **50** may be implemented in a virtual device such as a cloud server.

[0030] The information processing system **1** includes a snapshot information storage unit **20**, a process information storage unit **40**, a reference information storage unit **60**, a recognition result storage unit **70**, and a structured information storage unit **80**. Each of the storage units may be implemented by a storage device included in any one of the snapshot information generation device **10**, the process information generation device **30**, and the structured information generation device **50**. Each of the storage units may be implemented by a virtual device such as a cloud server.

Functional Configuration of Snapshot Information Generation Device 10

[0031] The snapshot information generation device **10** includes an information collection unit **110**, a status determination unit **120**, a meaning division unit **130**, and a snapshot information generation unit **140** as functional units.

[0032] The information collection unit **110** collects information indicating a state of an information collection subject. The information indicating the state of the information collection subject (hereinafter, also simply referred to as a subject) broadly includes information related to the state of the subject, such as a conversation, an utterance, a post on a social networking service (SNS), an electronic mail, a meeting minutes, health information obtained from wearable devices, purchasing history of goods and food, results of health checkups and medical tests of the subject, location information of the subject, a scattering state of fine particles such as pollen and air pollutants, and temperature, humidity, and air pressure conditions.

[0033] The information collection unit **110** may include a sound collection unit (microphone) that acquires audio information, an optical character reading unit (OCR) that acquires text information from paper media, a communication unit or storage medium reading unit that acquires text or audio information from other computer devices, or the like.

[0034] In addition to a content of the collected information (What), the information collection unit **110** also collects the information (so-called 5W1H) if there is information that identifies an individual sender (Who), and indicates a date and time when the information is sent (When), a place (Where), a situation (How), a reason for sending the information (Why), and the like.

[0035] Note that any one of the pieces of information of 5W1H may be missing (for example, some of information may be a Null value). For example, when the information to be collected is weather conditions around the subject, there

is no reason (Why) for the information (weather conditions) to occur. In this example, the information collection unit **110** collects information related to weather conditions in which the reason (Why) is missing among 5W1H.

[0036] The information collection unit **110** outputs the collected information to the status determination unit **120** and the meaning division unit **130**.

[0037] The status determination unit **120** determines a status in which the information collected by the information collection unit **110** is transmitted. If the information collected by the information collection unit **110** includes the information of 5W1H described above, the status determination unit **120** determines the status based on the information of 5W1H. For example, the information collected by the information collection unit **110** may include information (timing information) indicating a timing at which the state occurs and information (identification information) for identifying the subject. In this case, the status determination unit **120** determines that the information collected by the information collection unit **110** is information issued by the subject indicated by the identification information at the timing indicated by the timing information.

[0038] In the following description, the timing information and the identification information are also collectively referred to as situation information.

[0039] The status determination unit **120** outputs the situation information that is the determination result to the snapshot information generation unit **140**.

[0040] The meaning division unit **130** divides the information collected by the information collection unit **110** into units that have meaning in terms of content. The information divided by the meaning division unit **130** is also referred to as unit meaning information. For example, the meaning division unit **130** divides the information into unit meaning information by well-known natural language processing and automatic structuring. In the following description, the unit meaning information is also referred to as atom. The unit meaning information (atom) may be information divided into the smallest unit that has meaning in natural language.

[0041] The meaning division unit **130** outputs the unit meaning information to the snapshot information generation unit **140**.

[0042] The snapshot information generation unit **140** generates a snapshot information SSD by associating the situation information output by the status determination unit **120** with the unit meaning information output by the meaning division unit **130**.

[0043] The snapshot information generation unit **140** causes the snapshot information storage unit **20** to store the generated snapshot information SSD.

Functional Configuration of Process Information Generation Device 30

[0044] The process information generation device **30** includes a snapshot information acquisition unit **310** and a process information generation unit **320** as functional units thereof.

[0045] The snapshot information acquisition unit **310** acquires the snapshot information SSD from the snapshot information storage unit **20**. As described above, the snapshot information SSD is information in which the timing information and the situation information are associated with each other.

[0046] That is, the snapshot information acquisition unit **310** acquires the snapshot information SSD in which the unit meaning information and the situation information are associated with each other, here, the unit meaning information is obtained by dividing the information indicating the state of the subject into meaningful units, and the situation information includes at least the timing information indicating the timing at which the state indicated by the unit meaning information occurs and the identification information for identifying the subject.

[0047] Note that the information indicating the state of the subject described here includes information indicating the state of the subject himself or herself and information indicating a condition of an environment where the subject is located (for example, the weather and air pressure in a place where the subject is located). The environment where the subject is located does not matter directly or indirectly.

[0048] The snapshot information acquisition unit **310** outputs the acquired snapshot information SSD to the process information generation unit **320**.

[0049] The process information generation unit **320** generates process information PD in which the snapshot information SSD is arranged in a time series for each of the subjects based on the timing information.

[0050] The process information generation unit **320** causes the process information storage unit **40** to store the generated process information PD.

Functional Configuration of Structured Information Generation Device 50

[0051] The structured information generation device **50** includes a peculiarity determination unit **510**, a state information generation unit **520**, a structured information generation unit **530**, and a meaning imparting unit **540** as functional units thereof.

[0052] The peculiarity determination unit **510** acquires the process information PD stored in the process information storage unit **40** and reference information stored in the reference information storage unit **60**. That is, the peculiarity determination unit **510** functions as a process information acquisition unit that acquires the process information PD from the process information storage unit **40**.

[0053] The peculiarity determination unit **510** determines a peculiarity of the process information PD based on the process information PD and the reference information.

[0054] Here, the reference information is information indicating a standard (or reference) status among statuses indicated by the process information PD. The reference information is also referred to as master condition data.

[0055] The peculiarity determination unit **510** compares the process information PD stored in the process information storage unit **40** with the reference information, and determines whether the status indicated by the process information PD deviates from the standard status (or the reference status). When it is determined that the status indicated by the process information PD deviates from the standard status (or the reference status), the peculiarity determination unit **510** determines that the process information PD has a peculiarity.

[0056] The peculiarity determination unit **510** may determine the peculiarity of a part of the time series among the time series of the process information PD.

[0057] The peculiarity determination unit **510** determines a type of the peculiarity of the process information PD and groups the process information PD for each of the types of the peculiarity.

[0058] The state information generation unit **520** classifies the process information PD for the groups of the peculiarity determined by the peculiarity determination unit **510** to generate the state information CD for each of the groups.

[0059] The structured information generation unit **530** generates structured information MS by combining state information CD in different groups in the generated state information CD.

[0060] The meaning imparting unit **540** imparts a human-recognizable meaning to the state indicated by the structured information MS. Specifically, the meaning imparting unit **540** acquires a human recognition result stored in the recognition result storage unit **70**. The meaning imparting unit **540** compares the acquired recognition result with the state indicated by the structured information MS, and thereby imparting a human-recognizable meaning to the state indicated by the structured information MS.

[0061] Note that imparting a human-recognizable meaning to the state indicated by the structured information MS is also referred to as matching the structured information MS with the meaning. Information in which the human-recognizable meaning is imparted to the state indicated by the structured information MS is also referred to as a meaning matching structured information IMS. The meaning matching structured information IMS is also referred to as "identify-MAP (iMAP)".

[0062] The meaning imparting unit **540** causes the structured information storage unit **80** to store the meaning matching structured information IMS.

[0063] Next, a processing flow in which the process information generation device **30** and the structured information generation device **50** generate meaning matching structured information IMS based on the snapshot information SSD generated by the snapshot information generation device **10** will be described with reference to FIG. **2**.

Processing Flow in Recognition Mode

[0064] FIG. **2** is a diagram showing an example of a processing flow in the recognition mode of the information processing system **1** according to the present embodiment.

[0065] The snapshot information acquisition unit **310** acquires the snapshot information SSD from the snapshot information storage unit **20** (step S**110**). An example of the snapshot information SSD will be described with reference to FIG. **3**.

Example of Snapshot Information

[0066] FIG. **3** is a diagram showing an example of a structure of the snapshot information SSD according to the present embodiment. In the snapshot information SSD, a snapshot information ID, situation information, and unit meaning information are associated with each other.

[0067] The snapshot information SSD is divided for each of the subjects and at each of timings, and stored in the snapshot information storage unit **20**.

[0068] For example, each of snapshot information IDs (SS**11** to SS**13**) is the snapshot information SSD related to a subject U**01**. Each of snapshot information IDs (SS**21** to SS**23**) is snapshot information SSD related to a subject U**02**. Similarly, each of snapshot information IDs (SS**31** to SS**32**) is snapshot information SSD related to a subject U**03**.

[0069] In the snapshot information SSD related to the subject U**01**, the snapshot information ID (SS**11**) indicates unit meaning information M**01** which is a status generated at a timing T**01**.

[0070] Returning to FIG. **2**, the snapshot information acquisition unit **310** outputs the acquired snapshot information SSD to the process information generation unit **320**.

[0071] The process information generation unit **320** generates the process information PD based on the snapshot information SSD acquired in step S**110** (step S**120**). An example of the process information PD will be described with reference to FIG. **4**.

Example of Process Information

[0072] FIG. **4** is a diagram showing an example of a structure of the process information PD according to the present embodiment. The process information PD is information in which the snapshot information SSD is arranged in a time series for each of the subjects based on the timing information. (A) of FIG. **4** shows the process information PD of the subject U**01**. (B) of FIG. **4** shows the process information PD of the subject U**02**. (C) of FIG. **4** shows the process information PD of the subject U**03**.

[0073] More specifically, the process information generation unit **320** extracts the snapshot information SSD of a specific subject (for example, the subject U**01**) from the snapshot information SSD. The process information generation unit **320** arranges the extracted snapshot information SSD of the subject U**01** in a time series based on an occurrence timing indicated by the timing information. The process information generation unit **320** generates the snapshot information SSD of the subject U**01** arranged in a time series as the process information PD of the subject U**01**.

[0074] The process information generation unit **320** similarly generates the process information PD for other subjects (for example, the subject U**02**, the subject U**03**, . . .) by arranging the snapshot information SSD in a time series.

[0075] The process information generation unit **320** causes the process information storage unit **40** to store the generated process information PD of each of the subjects.

[0076] Returning to FIG. **2**, the peculiarity determination unit **510** determines the peculiarity of the process information PD (step S**130**).

[0077] Specifically, as described above, the peculiarity determination unit **510** compares the process information PD generated in step S**120** with the pre-stored reference information to determine whether the process information PD has a peculiarity.

[0078] Here, both the reference information and the process information PD are expressed in a format that can be compared by a computer (for example, a binarized data group). Therefore, it is difficult for a human to directly understand what the contents of the reference information and the process information PD indicate (for example, the meaning of information).

[0079] That is, the peculiarity determination unit **510** directly determines the peculiarity of the process information PD which has a data format that is difficult for a human to understand directly. In other words, the peculiarity determination unit **510** determines the peculiarity of the process information PD based on a data format that can be directly

compared by a computer, rather than based on a definition regarding the meaning of information by a human recognition (for example, ontology).

[0080] The state information generation unit **520** generates the state information CD by classifying the process information PD which is determined as information having a peculiarity in step S**130** into a plurality of groups for each type of the peculiarity (step S**140**). Here, a symptom of a cold is taken as an example, a peculiarity group includes a group with a symptom of cough, a group with a symptom of runny nose, a group with a symptom of sore throat, and the like. In this example, the peculiarity can be rephrased as a main complaint state of the subject. An example of the state information CD will be described with reference to FIG. **5**.

Example of State Information

[0081] FIG. **5** is a diagram showing an example of a structure of the state information CD according to the present embodiment. (A) of FIG. **5** shows state information CD**1** of a first group (for example, the group with the symptom of cough). (B) of FIG. **5** shows state information CD**2** of a second group (for example, the group with the symptom of runny nose). (C) of FIG. **5** shows state information CD**3** of a third group (for example, the group with the symptom of sore throat).

[0082] Returning to FIG. **2**, the structured information generation unit **530** generates the structured information MS by combining one or more pieces of the state information CD (step S**150**). The structured information MS described in the present embodiment may be expressed as meta structured information or super structured information. An example of the structured information MS will be described with reference to FIG. **6**.

Example of Structured Information

[0083] FIG. **6** is a diagram showing an example of a structure of the structured information MS according to the present embodiment.

[0084] (A) of FIG. **6** shows an example of structured information MS**1**. The structured information MS**1** is generated by combining the state information CD**1**, the state information CD**2**, and the state information CD**3** in the state information CD shown in FIG. **5**.

[0085] (B) of FIG. **6** shows an example of structured information MS**2**. The structured information MS**2** is generated by combining the state information CD**1** and the state information CD**2** in the state information CD shown in FIG. **5**.

[0086] (C) of FIG. **6 6** shows an example of structured information MS**3**. The structured information MS**3** is generated using the state information CD**3** in the state information CD shown in FIG. **5**.

[0087] Returning to FIG. **2**, the meaning imparting unit **540** generates the meaning matching structured information IMS by imparting the recognition information stored in the recognition result storage unit **70** to the structured information MS generated in step S**150** (step S**160**). An example of the meaning matching structured information IMS will be described with reference to FIG. **7**.

Example of Meaning Matching Structured Information

[0088] FIG. **7** is a diagram showing an example of a structure of the meaning matching structured information IMS according to the present embodiment. As described above, in this example, the structured information MS**1** is generated by combining the state information CD**1** (group: symptom of cough), the state information CD**2** (group: symptom of runny nose), and the state information CD**3** (group: symptom of sore throat). In the case of the combination of symptoms, if a diagnosis name A or a diagnosis name B applies thereto, the recognition result storage unit **70** stores the recognition result based on the information to be analyzed (for example, communication information described later). In this case, the meaning imparting unit **540** imparts the diagnosis name A or the diagnosis name B to the structured information MS**1**.

[0089] Similarly, the meaning imparting unit **540** imparts a diagnosis name C to the structured information MS**2**, and imparts a diagnosis name D to the structured information MS**3**.

[0090] Returning to FIG. **2**, the meaning imparting unit **540** causes the structured information storage unit **80** to store the meaning matching structured information IMS to which the recognition information is imparted, and ends the series of processing.

Configuration of Information Processing System **1** in Application Mode

[0091] Next, the configuration of the information processing system **1** in the application mode will be described with reference to FIG. **8**. The same configurations as those of the information processing system **1** described in the above recognition mode are denoted by the same reference numerals, and descriptions thereof are omitted.

[0092] FIG. **8** is a diagram showing an example of the configuration of the information processing system **1** in the application mode according to the present embodiment. In the application mode, the information processing system **1** differs from the configuration in the recognition mode in that the information processing system **1** includes a process complement device **90**.

[0093] The process complement device **90** includes a communication information acquisition unit **910**, an information supply unit **920**, an extraction unit **930**, and a presentation unit **940** as functional units thereof.

[0094] The communication information acquisition unit **910** acquires communication information indicating contents of communication between users.

[0095] The extraction unit **930** extracts, based on the acquired communication information and the structured information MS, information required to be complemented based on the structured information MS as complementary information in a time series of the communication.

[0096] The presentation unit **940** presents the extracted complementary information to the user.

[0097] The information supply unit **920** supplies information indicating the state of the subject obtained as a result of the presentation of the complementary information presented by the presentation unit **940** to the snapshot information acquisition unit **310**.

[0098] A more specific processing flow in the application mode will be described with reference to FIG. **9**.

Flow in Application Mode

[0099] FIG. **9** is a diagram showing an example of a processing flow in the application mode of the present embodiment. In this example, a case where the information processing system **1** is used as an examination navigation system during the examination of a patient by a doctor will be described. In this case, the user of the information processing system **1** includes a doctor and a patient.

[0100] The communication information acquisition unit **910** acquires communication information (step S**210**). In this example, the communication information acquisition unit **910** includes, for example, a sound collection unit (microphone) and an information acquisition unit from an electronic medical record system. The communication information acquisition unit **910** acquires a conversation between the doctor and the patient during the examination, a description content of the electronic medical record, and the like as communication information.

[0101] The communication information acquisition unit **910** outputs the acquired communication information to the snapshot information generation device **10** (step S**220**). The snapshot information generation device **10** generates the snapshot information SSD based on the communication information. The snapshot information storage unit **20** generates the process information PD based on the generated snapshot information SSD. The process information PD generated in step S**220** indicates a process of a content of communication between the doctor and the patient in the application mode (that is, a content exchanged in real time).

[0102] The extraction unit **930** acquires the process information PD generated by the snapshot information storage unit **20** in step S**220** (that is, the process information PD generated in real time) (step S**230**). The extraction unit **930** compares the process information PD generated in the real time with the meaning matching structured information IMS stored in the structured information storage unit **80** (that is, the meaning matching structured information IMS generated in the recognition mode). The extraction unit **930** extracts information to be complemented based on the structured information MS as complementary information in time series of communication based on the result of the comparison.

[0103] For example, in correspondence with information to which the computer-recognizable or the human-recognizable meaning is imparted, coincidence is significantly improved if there is a blood glucose level, and in this case, if the communication information exchanged between the doctor and the patient during the examination does not include information related to the blood glucose level (inspection value), the blood glucose level (inspection value) is requested.

[0104] For example, it is assumed that the doctor and the patient during the examination have a conversation about symptoms such as "I started coughing yesterday", "I went to sleep with coughing last night", and "I started having a runny nose this morning".

[0105] In this case, in step S**220**, the snapshot information SSD**1** including the timing T**01** "yesterday" and the unit meaning information M**01** "I started coughing", and the snapshot information SSD**2** including the timing T**03** "this morning" and the unit meaning information M**03** "I started having a runny nose" are generated.

[0106] Further, in step S**220**, the process information PD is generated based on the snapshot information SSD**1** and the snapshot information SSD**2**. The process information PD indicates the passage of the symptoms such as "I started coughing yesterday" (=timing T**01**, unit meaning information M**01**) and "I started having a runny nose this morning" (=timing T**03**, unit meaning information M**03**).

[0107] The extraction unit **930** searches the meaning matching structured information IMS stored in the structured information storage unit **80** using the generated process information PD as a search key. As a result of this search, for example, in the case of an example of the meaning matching structured information IMS shown in FIG. **7**, "timing T**01**, unit meaning information M**01**", and "timing T**03**, and unit meaning information M**03**" of meaning matching structured information IMS**1** shown in (A) of FIG. **7** are hit.

[0108] The extraction unit **930** acquires the meaning matching structured information IMS**1** as information of a complement source. The extraction unit **930** extracts, as complementary information, information associated with timings T**04** to T**08** following the timing T**03** in the time series of communication indicated by the meaning matching structured information IMS**1**. For example, the extraction unit **930** extracts the diagnosis name A and the diagnosis name B shown in (A) of FIG. **7** as the complementary information.

[0109] In the case of the example described above, the "timing T**01**, unit meaning information M**01**" and the "timing T**03**, unit meaning information M**03**" of meaning matching structured information IMS**2** shown in (A) of FIG. **7** are also hit as information tracing the passage of the "timing T**01**, unit meaning information M**01**" and the "timing T**03**, and the unit meaning information M**03**".

[0110] In this case, the extraction unit **930** also acquires the meaning matching structured information IMS**2** as the information of the complement source. The extraction unit **930** extracts information associated with the timing T**04** to T**05** following the timing T**03** as the complementary information in a time series of communication indicated by the meaning matching structured information IMS**2**. For example, the extraction unit **930** extracts the diagnosis name C shown in (B) of FIG. **7** as the complementary information.

[0111] In the case of the example described above, it is assumed that the doctor and the patient further continue the conversation and the patient says that "My throat is starting to hurt now". In this case, the process information PD including "timing T**07**, unit meaning information M**07**" and corresponding to "My throat is starting to hurt now" is generated. When the newly generated process information PD is added, the meaning matching structured information IMS**1** is hit, but the meaning matching structured information IMS**2** is not hit.

[0112] In this case, the extraction unit **930** may extract the complementary information by removing the meaning matching structured information IMS**2** from candidates of the complement source and using the meaning matching structured information IMS**1** as the information of the complement source. For example, the extraction unit **930** excludes the diagnosis name C shown in (B) of FIG. **7**, and extracts the diagnosis name A or the diagnosis name B shown in (A) of FIG. **7** as the complementary information.

[0113] Returning to FIG. **9**, the presentation unit **940** presents a user (in this case, a doctor) with the complementary information extracted by the extraction unit **930** (step S**240**). Specifically, the presentation unit **940** is connected

to, for example, a display device **2** including a liquid crystal display. In this case, the presentation unit **940** causes the display device **2** to display the complementary information (for example, the diagnosis name A or the diagnosis name B) extracted by the extraction unit **930**.

[0114] The communication information acquisition unit **910** acquires, as the communication information, information indicating the state of the subject obtained as a result of the presentation of the complementary information presented by the presentation unit **940** (for example, results of further consultation with the doctor based on the complementary information) (step S**250**).

[0115] When a condition for providing the communication information is satisfied, for example, in a case where a consent is obtained from a patient, the information supply unit **920** supplies the communication information to the snapshot information generation device **10** as a generation source information of new snapshot information SSD.

[0116] As a result, the snapshot information SSD stored in the snapshot information storage unit **20** can be updated with new information about the conversation between the doctor and the patient. According to such a configuration, more snapshot information SSD can be collected. According to the information processing system **1** configured in this way, accuracy of the meaning matching structured information IMS can be further improved.

[0117] As described above, the information processing system **1** according to the present embodiment includes the process information generation device **30** and the structured information generation device **50**, and generates the structured information MS from the snapshot information SSD.

[0118] On the one hand, in the related art, it has been proposed to generate a database for work support based on a definition regarding a meaning of information by a human recognition. However, in a case where the database is generated based on the definition by the human recognition, there is a problem in that work of imparting the human recognition to information is necessary, and an enormous number of man-hours are required.

[0119] According to the information processing system **1** of the present embodiment, the structured information MS is generated in the data format on a computer, which is difficult for a human to directly understand, so that the structured information MS can be generated without the work of imparting the human recognition to the information.

[0120] Therefore, the information processing system **1** can generate the structured information MS based on an enormous amount of snapshot information SSD.

[0121] Therefore, according to the information processing system **1** of the present embodiment, it is also possible to generate the structured information MS by using, as the snapshot information SSD, information that a human is not aware of, or information that a human is aware of but difficult to capture because the amount of information is too large. Therefore, according to the information processing system **1** of the present embodiment, the amount and accuracy of the structured information MS can be improved, and more appropriate complementary information can be presented.

[0122] The information processing system **1** according to the present embodiment generates the meaning matching structured information IMS in which the human recognition is imparted to the structured information MS. According to the information processing system **1** configured as described above, even if the structured information MS is constructed in a data format that is difficult for a human to directly understand, information can be provided in a human-recognizable format.

[0123] Therefore, according to the information processing system **1** of the present embodiment, it is possible to simultaneously improve the amount and accuracy of the structured information MS based on an enormous amount of snapshot information SSD and present the content of the structured information MS in a human-understandable format.

[0124] Although embodiments of the present invention have been described in detail above with reference to the drawings, the specific configuration is not limited to these embodiments, and appropriate modifications can be made without departing from the gist of the present invention. The configurations described in the above embodiments may be combined.

[0125] Each of the units of each of the devices in the above embodiments may be implemented by dedicated hardware, or may be implemented by a memory and a microprocessor.

[0126] Each of the units of each of the devices may be configured by a memory and a central processing unit (CPU), and may implement a function by loading a program for implementing a function of the unit of the device into the memory and executing the program.

[0127] A program for implementing the function of each of the units of each of the devices may be recorded on a computer-readable recording medium, and the program recorded on this recording medium may be read into a computer system and executed to perform processing by each of the units of the control unit. The "computer system" herein includes an OS and hardware such as a peripheral device.

[0128] Further, the "computer system" includes a home page providing environment (or display environment) when a WWW system is used.

[0129] The "computer-readable recording medium" refers to a storage device, for example, a portable medium such as a flexible disk, a magneto-optical disk, a ROM, and a CD-ROM, and a hard disk built in the computer system. Further, the "computer-readable recording medium" includes one that dynamically holds a program for a short period of time, such as a communication line when transmitting a program via a network such as the Internet or a communication line such as a telephone line, and one that holds a program for a certain period of time, such as a volatile memory in a computer system serving as a server or a client in that case. The above program may be for implementing a part of the functions described above, and may further be capable of implementing the functions described above in combination with a program already recorded in the computer system.

Reference Signs List

[0130] **1** information processing system
[0131] **10** snapshot information generation device
[0132] **20** snapshot information storage unit
[0133] **30** process information generation device
[0134] **40** process information storage unit
[0135] **50** structured information generation device
[0136] **60** reference information storage unit
[0137] **70** recognition result storage unit

[0138] **80** structured information storage unit

[0139] **90** process complement device

1. An information processing system, comprising:

a snapshot information acquisition unit configured to acquire snapshot information in which unit meaning information and situation information are associated with each other, the unit meaning information being obtained by dividing information indicating a state of a subject into a meaningful unit, the situation information including at least timing information indicating a timing at which the state indicated by the unit meaning information occurs and identification information for identifying the subject; and

a process information generation unit configured to generate process information in which the snapshot information is arranged in a time series for each of the subjects based on the timing information.

2. The information processing system according to claim **1**, further comprising:

a peculiarity determination unit configured to determine a peculiarity of the process information generated by the process information generation unit based on the process information and reference information indicating a predetermined reference;

a state information generation unit configured to classify the process information for groups of the peculiarity determined by the peculiarity determination unit to generate state information for each of the groups; and

a structured information generation unit configured to generate structured information by combining the state information in different groups in the generated state information.

3. The information processing system according to claim **2**, further comprising:

a meaning imparting unit configured to impart a meaning to the state indicated by the structured information.

4. The information processing system according to claim **3**, further comprising:

a communication information acquisition unit configured to acquire communication information indicating a content of communication between users;

an extraction unit configured to extract, based on the acquired communication information and the structured information, information, which is required to be complemented based on the structured information, in a time series of the communication as complementary information; and

a presentation unit configured to present the extracted complementary information to the users.

5. The information processing system according to claim **4**, further comprising:

an information supply unit configured to supply, to the snapshot information acquisition unit, information indicating a state of a subject obtained as a result of the presentation of the complementary information presented by the presentation unit.

6. An information processing system, comprising:

a process information acquisition unit configured to acquire process information in which snapshot information, in which unit meaning information obtained by dividing information indicating a state of a subject into a meaningful unit and situation information are associated with each other, is arranged in a time series for each of the subjects based on timing information indicating a timing at which the state indicated by the unit meaning information occurs, the situation information including at least the timing information and identification information for identifying the subject;

a peculiarity determination unit configured to determine a peculiarity of the acquired process information based on the process information and reference information indicating a predetermined reference;

a state information generation unit configured to classify the process information for groups of the peculiarity determined by the peculiarity determination unit to generate state information for each of the groups; and

a structured information generation unit configured to generate structured information by combining the state information in different groups in the generated state information.

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